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MatAnyone: Stable Video Matting with Consistent Memory Propagation

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https://pq-yang.github.io/projects/MatAnyone



Figure 1. Our *MatAnyone* is capable of producing highly detailed and temporally consistent alpha mattes throughout a video. (a) It adapts to a variety of frame sizes and media types (*e.g.*, films, games, smartphone videos), achieving fine-grained details at the image-matting level. (b) RVM [30], an auxiliary-free video matting method, struggles with complex or ambiguous backgrounds. In contrast, our method effectively isolates the target object from such distractors, preserving a clean background and complete foreground parts. (c) Our method also excels at consistently tracking the target (*i.e.*, the lady in pink) even in scenes containing multiple salient objects (*i.e.*, the man and the lady). It accurately distinguishes between them even during their interactions. **(Zoom-in for best view)**

Abstract

Auxiliary-free human video matting methods, which rely solely on input frames, often struggle with complex or ambiguous backgrounds. To address this, we propose MatAnyone, a robust framework tailored for target-assigned video matting. Specifically, building on a memory-based paradigm, we introduce a consistent memory propagation module via region-adaptive memory fusion, which adaptively integrates memory from the previous frame. This ensures semantic stability in core regions while preserving fine-grained details along object boundaries. For robust training, we present a larger, high-quality, and diverse dataset for video matting. Additionally, we incorporate a novel training strategy that efficiently leverages large-scale segmentation data, boosting matting stability. With this new

network design, dataset, and training strategy, MatAnyone delivers robust and accurate video matting results in diverse real-world scenarios, outperforming existing methods.

1. Introduction

Auxiliary-free human video matting (VM) is widely recognized for its convenience [21, 24, 30], as it only requires input frames without additional annotations. However, its performance often deteriorates in complex or ambiguous backgrounds, especially when similar objects, *i.e.*, other humans, appear in the background (Fig. 2(b)). We consider auxiliary-free video matting to be under-defined, as their results can be uncertain without a clear target object.

In this work, we focus on a problem that is more applicable to real-world video applications: video matting focused on pre-assigned target object(s), with the target segmentation mask provided in the first frame. This enables the model to perform stable matting via consistent object tracking throughout the entire video, while offering better interactivity. The setting is well-studied in Video Object Segmentation (VOS), where it is referred to as "semi-supervised" [10, 17, 34]. A common strategy is to use a memory-based paradigm [8, 12, 34, 46], encoding past frames and corresponding segmentation results into memory, from which a new frame retrieves relevant information for its mask prediction. This allows a lightweight network to achieve consistent and accurate tracking of the target object. Inspired by this, we adapt the memory-based paradigm for video matting, leveraging its stability across frames.

Video matting poses additional challenges compared to VOS, as it requires not only *accurate semantic detection* in *core* regions but also *high-quality detail extraction* along the *boundary* (*e.g.*, hair), as defined in Fig. 2(a). A straightforward approach is to fine-tune matting details using matting data, based on segmentation priors from VOS. Recent approaches attempt to achieve both goals, either in a coupled or decoupled manner. For instance, AdaM [28] and FTP-VM [19] refine the memory-based segmentation mask for each frame via a decoder to produce alpha mattes, while MaGGIe [20] devises a separate refiner network to process segmentation masks across all frames from an off-the-shelf VOS model. However, these methods often lead to suboptimal results due to limitations in the available video matting data: (i) the quality of VideoMatte240K [29], the most widely used human video matting dataset, is suboptimal. Its ground-truth alpha mattes exhibit problematic semantic accuracy in core areas (*e.g.*, interior holes) and lack fine details along the boundaries (*e.g.*, blurry hair); (ii) video matting datasets are much smaller in scale compared to VOS datasets; and (iii) video matting data are synthetic due to the extreme difficulty of human annotations, limiting their generalizability to real-world cases [30]. Consequently, fine-tuning a strong VOS prior for video matting with existing video matting data usually disrupts this prior. While boundary details may show improvement compared to segmentation results, the matting quality in terms of semantic stability in *core* areas and details in *boundary* areas remain unsatisfactory, as shown by the results of MaGGIe in Fig. 2(b).

Producing matting-level details while maintaining semantic stability of a memory-based approach is challenging, especially training with suboptimal video matting data. To tackle this, we focus on several key aspects:

Network - we introduce a *consistent* memory propagation mechanism in the memory space. For each current frame, the alpha value change relative to the previous frame is estimated for every token. This estimation guides the adaptive integration of information from the previous frame. The "large-change" regions rely more on the current frame's information queried from the memory bank, while "small-change" regions tend to retain the memory from the previous frame. This region-adaptive memory fusion inherently stabilizes memory propagation throughout the video, improving matting quality with fine details and temporal consistency. Specifically, it encourages the network to focus on boundary regions during training to capture fine details, while "small-change" tokens in the core regions preserve internally complete foreground and clean background (see our results in Fig. 2(b)).

Data - we collect a new training dataset, named *VM800*, which is twice as large, more diverse, and of higher quality in both core and boundary regions compared to the VideoMatte240K dataset [29], greatly enhancing robust training for video matting. In addition, we introduce a more challenging test dataset, named *YoutubeMatte*, featuring more diverse foreground videos and improved detail quality. These new datasets offer a solid foundation for robust training and reliable evaluation in video matting.

Training Strategy - the lack of *real* video matting data remains a significant limitation, affecting both stability and generalizability. We address this problem by leveraging large-scale *real* segmentation data via a novel training strategy. Unlike common practices [19, 20, 30] that train with segmentation data on a separate prediction head parallel to the matting head, we propose using segmentation data within the same head as matting for more effective supervision. This is achieved by applying region-specific losses – for *core* regions, we apply a pixel-wise loss to ensure stability and generalization in semantics; for *boundary* regions, where segmentation data lacks alpha labels, we employ an improved DDC loss [32], scaled to make edges resemble matting rather than segmentation.

In summary, our main contributions are as follows:

- We propose *MatAnyone*, a practical human video matting framework supporting target assignment, with stable performance in both semantics of *core* regions and fine-grained *boundary* details. Target object(s) can be easily assigned using off-the-shelf segmentation methods, and reliable tracking is achieved even in long videos with

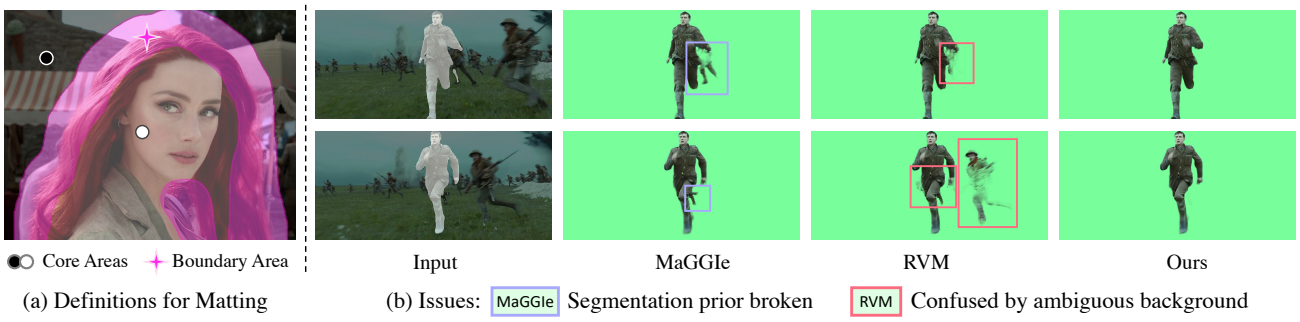


Figure 2. Definitions and motivations for MatAnyone. (a) In a matting frame, the image can be broadly divided into two areas based on the alpha value: the *core* (semantic) and the *boundary* (fine-details). The *core* includes the background (alpha values of 0) and the solid foreground (alpha values of 1), while the *boundary* (highlighted in pink) encompasses areas with alpha values between 0 and 1. (b) Due to the under-defined setting, auxiliary-free methods like RVM [30] are easily confused by ambiguous background. Meanwhile, mask-guided methods like MaGGIe [20] tend to break the segmentation prior they aim to leverage, due to the deficiency in video matting data.

complex and ambiguous backgrounds.

- We introduce a *consistent* memory propagation mechanism via *region-adaptive* memory fusion, improving stability in *core* regions and quality in *boundary* details.
- We contribute larger and higher-quality datasets for training and testing, offering a solid foundation for robust training and reliable evaluation in video matting.
- To overcome the scarcity of real video matting data, we leverage real segmentation data for core-area supervision, largely improving semantic stability over prior methods.

2. Related Work

Video Matting. Due to the intrinsic ambiguity in the auxiliary-free setting [21, 24, 30, 35, 51, 56], such tasks generally are object-specific. Among them, human video matting [21, 24, 39, 56] without auxiliary inputs is popular due to its wide applications. Challenging as the auxiliary-free setting, being in the video domain brings in additional difficulties in temporal coherency. MODNet [21] extends its portrait matting setting to video domain with a flickering reduction trick (non-learning) within a local sequence. RVM [30] steps further to design for videos specifically with ConvGRU [1] as its recurrent architecture. Robust as RVM, it is still easy to be confused by humans in the background. With the success of promptable segmentation [22, 36, 52, 57], obtaining segmentation mask for a target human object only requires minimal human efforts. Recent mask-guided image [3, 26, 49, 50] and video matting [19, 20, 25, 28] thus leverages this convenience for a more robust performance. Adam [28] propagates the first-frame segmentation mask across all frames while FTP-VM [19] propagates the first-frame trimap. Taking the propagated mask as a rough result, their decoder serves for matting details refinement. MaGGIe [20] enjoys a stronger prior by taking the segmentation mask across all frames instead of the first one. Taking all the segmentation masks at a time, the network is able to perform bidirectional temporal fusion for coherency. To mitigate the poor generalizability of synthetic video matting data, a common practice is to simultaneously train with real segmentation data for semantics supervision [19, 28, 30].

Memory-based VOS. Semi-supervised VOS segments the target object with a first-frame annotation across frames [8–12, 16, 27, 33, 38]. The memory matching paradigm by Space-Time Correspondence Network (STCN) [10] is widely followed by current VOS methods [8, 12, 41, 46], and achieves good performance. We thus take the memory-based paradigm as our framework since it is similar to our setting except that our outputs are alpha mattes.

Video Consistency in Low-level Vision. To enhance temporal consistency across adjacent frames, the recurrent frame fusion [42, 53] and optical flow-guided propagation [4–6, 54] are commonly utilized in the video restoration networks. Recent methods also employ temporal layers such as 3D convolution [2, 43] and temporal attention [2, 7, 44, 55] during training, while other training-free methods resorts to cross-frame attention [45, 48] and flow-guided attention [13, 15] in the pretrained models. In this work, we find that the memory-based paradigm is effective enough to maintain video consistency for video matting.

3. Methodology

Overview. Achieving matting-level details while preserving the semantic stability of a memory-based approach poses challenges, especially when training with suboptimal video matting data. To tackle this, we propose our *MatAnyone*, as illustrated in Fig. 3. Similar to semi-supervised VOS, MatAnyone only requires the segmentation mask for the first frame as a target assignment (*e.g.*, the yellow mask in Fig. 3(a)). The alpha matte for the assigned object is then generated frame by frame in a sequential manner. Specifically, for an incoming frame t, it is first encoded into F^t as $\times 16$ downsampled feature representation, which is then transformed into key and query for consistent memory propagation (Sec. 3.1), and output the pixel memory readout P^t. We employ the object transformer proposed by Cutie [12] to group the pixel memory by object-level semantics for robustness against noise brought by low-level pixel matching.

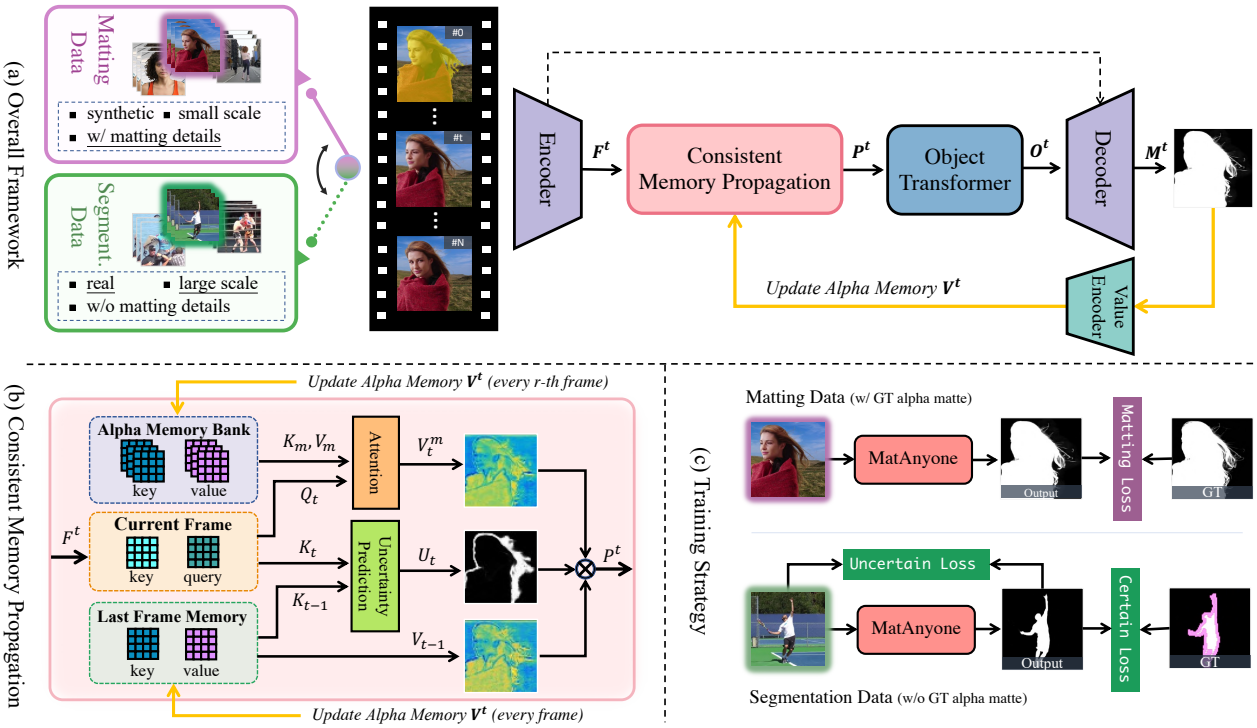


Figure 3. An overview of MatAnyone. MatAnyone is a memory-based framework for video matting. Given a target segmentation map in the first frame, our model achieves stable and high-quality matting through consistent memory propagation, with a region-adaptive memory fusion module to combine information from the previous and current frame. To overcome the scarcity of real video matting data, we incorporate a new training strategy that effectively leverages matting data for fine-grained matting details and segmentation data for semantic stability, with designed losses separately.

The refined memory readout O^t acts as the final feature to be sent into the decoder for alpha matte prediction. The predicted alpha matte M^t is then encoded to memory value V^t, which is used to update the alpha memory bank.

Due to limitations in the quality and quantity of video matting data, training with such data makes it difficult to achieve satisfactory stability in core regions. To mitigate this, RVM [30] proposes a parallel head for *real* segmentation data alongside the matting head, guiding the network to be robust in real-world cases. However, this is not sufficient, as the matting head itself cannot receive supervision from real data. Inspired by the DDC loss [32] designed for alpha-free image matting, we devise a training strategy for core regions, which provides *direct* supervision to the matting head with segmentation data (Sec. 3.2), leading to substantial improvements in semantic stability.

We also propose a practical inference strategy that allow for flexible application: a recurrent refinement approach applied to the first frame, based on the memory-driven paradigm, enhancing robustness to the given mask and refining matting details (Sec. 3.3).

3.1. Consistent Memory Propagation

Alpha Memory Bank. In this study, we introduce a *consistent memory propagation* (CMP) module specifically designed for video matting, as illustrated in Fig.3(b). Existing memory-based VM methods store either segmentation masks [28] or trimaps [19] in memory and use a decoder to refine the matting details. Such approaches do not fully leverage the stability provided by the memory paradigm in boundary regions, leading to instability such as flickering. To address this, building on the memory-based framework [10], our MatAnyone stores the alpha matte in an alpha memory bank to enhance stability in boundary regions.

Region-Adaptive Memory Fusion. Given the inherent difference between the segmentation map (values of 0 or 1) and the matting map (values between 0 and 1), the memory-matching approach needs to be adjusted. Specifically, in STCN [10], memory values for the query frame are based on the similarity between query and memory key, assuming equal importance for all query tokens. However, this assumption does not hold for video matting. As shown in Fig. 2(a), a query frame can be divided into *core* and *boundary* regions. When compared with frame $t-1$, only a small fraction of tokens in frame t change significantly in alpha values, with these "large-change" tokens mainly located in object boundaries, while the "small-change" tokens reside in the core regions. This highlights the need to treat *core* and *boundary* regions separately to enforce stability.

Specifically, we introduce a boundary-area prediction module to estimate the change probability U_t of each query token for adaptive memory fusion, where higher U_t indi-

cates "large-change" regions and lower U_t indicates "small-change" regions. The prediction module is lightweight, consisting of three convolution layers. We formulate the prediction as a binary segmentation problem with loss $\mathcal{L}_{bin_seg}$ and use the actual alpha change between frame $t-1$ and t as supervision. Specifically, we define U_t^{GT} : $|M_{t-1}^{GT} - M_t^{GT}| >= \delta$, where δ is a threshold. Using the output of the module $\hat{U}_t$, we compute the binary cross entropy loss against U_t^{GT}. During the region-adaptive memory fusion process, we apply the sigmoid function on $\hat{U}_t$ to transform it as a probability. The final pixel memory readout is a soft merge:

$$P_t = V_t^m * U_t + V_{t-1} * (1 - U_t), \tag{1}$$

where $U_t \in [0, 1]$, V_t^m are current values queried from memory bank, and V_{t-1} are values propagated from the last frame. This approach significantly improves stability in core regions by maintaining internal completeness and a clean background (Fig. 2(b) and Fig. 4). It also enhances stability in boundary regions, as it directs the network to focus on object boundaries with soft alpha values, while the memory-based paradigm inherently stabilizes the matched values (see Table 3(c)). A detailed analysis is provided in the ablation study of Sec. 5.2 and Sec. D.2 (supplementary).

3.2. Core-area Supervision via Segmentation

New Training Scheme. Most recent video matting methods follow RVM's approach of using real segmentation data to address the limitations of video matting data. In these methods, segmentation and matting data are fed to the main shared network, but are directed to produce outputs at separate heads. Although segmentation data do supervise the main network to empower generalizability and robustness to the matting model, the stability they provide falls short of what a VOS model could achieve. As shown in Fig. 2, both RVM and MaGGIe perform significantly worse than the VOS outputs (white masks on inputs) by XMem [8] in core areas, where semantic information is key. We believe the parallel head training scheme may not fully exploit the rich segmentation prior in the data. To address this, we propose to supervise the matting head directly with segmentation data. Specifically, we predict the *alpha matte* for segmentation inputs and optimize the matting outputs accordingly, as illustrated in Fig. 3(c).

Scaled DDC Loss. A natural challenge arises with the aforementioned approach: how can we compute the loss on matting outputs for segmentation data when there is no ground truth (GT) alpha matte? For core areas, the GT labels are readily available in the segmentation data, where an $l1$ loss suffices, and we denote it as $\mathcal{L}_{core}$. The real difficulty lies in the boundary region. A recent paper proposes DDC loss [32], which supervises boundary areas using the input image without requiring a GT alpha matte.

$$\mathcal{L}_{DDC} = \tfrac{1}{N}\sum_{i}^{N}\sum_{j} |\alpha_i - \alpha_j - \|\boldsymbol{I}_i - \boldsymbol{I}_j\|_2|, \quad j \in \text{argtopk}\{-\|\boldsymbol{I}_i - \boldsymbol{I}_j\|_2\}. \tag{2}$$

However, we find that the underlying assumption of this design, that $\|\alpha_i - \alpha_j\|_2 = \|\boldsymbol{I}_i - \boldsymbol{I}_j\|_2$ for $\alpha_i > \alpha_j$, does not always hold true. For two image pixels $\boldsymbol{I}_i$ and $\boldsymbol{I}_j$, their difference is given by:

$$\boldsymbol{I}_i - \boldsymbol{I}_j = [\alpha_i F_i + (1-\alpha_i)B_i] - [\alpha_j F_j + (1-\alpha_j)B_j], \tag{3}$$

where F_i, B_i represent the foreground and background values at pixel i, and similarly for F_j and B_j at pixel j. Since we impose the constraint $j \in \text{argtopk}\{-\|\boldsymbol{I}_i - \boldsymbol{I}_j\|_2\}$, we can assume $F_i = F_j = F$, $B_i = B_j = B$ within a small window. This simplifies Eq. (3) to:

$$\boldsymbol{I}_i - \boldsymbol{I}_j = (\alpha_i - \alpha_j)(F - B). \tag{4}$$

This shows that the assumptions for DDC loss hold only when $|F - B| = 1$. To account for this, we devise a scaled version as our boundary loss $\boldsymbol{L}_{boundary}$:

$$\mathcal{L}_{boundary} = \tfrac{1}{N}\sum_{i}^{N}\sum_{j} |(\alpha_i - \alpha_j)(\mathbf{F} - \mathbf{B}) - \|\boldsymbol{I}_i - \boldsymbol{I}_j\|_2|, \quad j \in \text{argtopk}\{-\|\boldsymbol{I}_i - \boldsymbol{I}_j\|_2\}, \tag{5}$$

where F is approximated by the average of the top k largest pixel values in the small window, and B by the average of the top k smallest pixel values. In the ablation study (Sec. 5.2), we show that training with our scaled DDC loss (Eq. (5)) yields more natural matting results than training with the original version (Eq. (2)), which tends to produce segmentation-like jagged and stair-stepped edges.

3.3. Recurrent Refinement During Inference

The first-frame matte is predicted from the given first-frame segmentation mask, and its quality will affect the matte prediction for the subsequent frames. The sequential prediction in the memory-based paradigm enables recurrent refinement during inference. Leveraging this mechanism, we introduce an optional first-frame warm-up module for inference. Specifically, we repeat the first frame n times, treating each repetition as the initial frame, and use only the nth alpha output as the first frame to initialize the alpha memory bank. This (1) enhances robustness against the given segmentation mask and (2) refines matting details in the first frame to achieve image-matting quality (see Fig. 6 and Fig. E in the supplementary).

4. Data

We briefly introduce our new training datasets and benchmarks for evaluation, including both synthetic and real-world. More details are provided in the appendix (Sec. C).

Table 1. Quantitative comparisons on different video matting benchmarks from diverse sources. The best and second-best performances are marked in red and orange, respectively. † indicates that MaGGIe [20] requires the instance mask as guidance for each frame, while our method only requires it in the first frame.

Metrics	Auxiliary-free (AF) Methods			Mask-guided Methods			
	MODNet [21]	RVM [30]	RVM-Large [30]	AdaM [28]	FTP-VM [18]	MaGGIe† [20]	**Ours**
VideoMatte (512 × 288)							
MAD↓	9.41	6.08	5.32	5.30	6.13	5.49	5.15
MSE↓	4.30	1.47	0.62	0.78	1.31	0.60	0.93
Grad↓	1.89	0.88	0.59	0.72	1.14	0.57	0.67
dtSSD↓	2.23	1.36	1.24	1.33	1.60	1.39	1.18
Conn↓	0.81	0.41	0.30	0.30	0.41	0.31	0.26
VideoMatte (1920 × 1080)							
MAD↓	11.13	6.57	5.81	4.42	8.00	4.42	4.24
MSE↓	5.54	1.93	0.97	0.39	3.24	0.40	0.33
Grad↓	15.30	10.55	9.65	5.12	23.75	4.03	4.00
dtSSD↓	3.08	1.90	1.78	1.39	2.37	1.31	1.19
YoutubeMatte (512 × 288)							
MAD↓	19.37	4.08	3.36	-	3.08	3.54	2.72
MSE↓	16.21	1.97	1.04	-	1.29	1.23	1.01
Grad↓	2.05	1.34	1.03	-	1.16	1.10	0.97
dtSSD↓	2.79	1.81	1.62	-	1.83	1.88	1.60
Conn↓	2.68	0.60	0.50	-	0.41	0.49	0.39
YoutubeMatte (1920 × 1080)							
MAD↓	15.29	4.37	3.58	-	6.49	2.37	1.99
MSE↓	12.68	2.25	1.23	-	4.58	0.98	0.71
Grad↓	8.42	15.1	12.97	-	29.78	7.69	8.91
dtSSD↓	2.74	2.28	2.04	-	2.41	1.77	1.65

4.1. Training Datasets

To address limitations in video matting datasets in both quality and quantity, we collect abundant green screen videos, process them with Adobe After Effects, and conduct manual selection to remove common artifacts also found in VideoMatte240K [29] (see Fig. A). Compared to VideoMatte240K, our dataset, *VM800*, is *(1)* twice as large, *(2)* more diverse in terms of hairstyles, outfits, and motion, and *(3)* higher in quality. Ablation studies (Table 3(b) and Sec. D.1) further demonstrate the advantages of our dataset.

4.2. Synthetic Benchmark

The standard benchmark, VideoMatte [29], derived from VideoMatte240K, includes only *5* unique foreground videos, which is under representative. Additionally, their foregrounds lack sufficient boundary details, limiting their ability to discern matting precision in boundary regions. To create a more comprehensive benchmark, we compile *32* distinct 1920 × 1080 green-screen foreground videos from YouTube, and process them similarly to our training dataset. Our benchmark, *YouTubeMatte*, provides enhanced detail representation, as reflected by higher Grad [37] values.

4.3. Real-world Benchmark and Metric

Real-world benchmarks are essential to facilitate the practical use of video matting models. Although real-world videos lack ground truth (GT) alpha mattes, we can generate frame-wise segmentation masks as GT for *core* areas benefiting from the high capability of existing VOS methods. Specifically, we select a subset of 25 real-world videos [30] (100 frames each) with high-quality core GT masks verified manually. MAD, MSE, and dtSSD [14] are then calculated at the core region as core region metrics, representing semantic stability that is critical for visual perception.

5. Experiments

Training Schedule. *Stage 1.* Following the practice of RVM [30], we start by training the entire model on our VM800 for 80k iterations. The sequence length is initially set to 3 and extended to 8 with increasing sampling intervals for more complex scenarios. *Stage 2.* As the key stage, we apply the core supervision training strategy introduced in Section 3.2. Real segmentation data COCO [31], SPD [40] and YouTubeVIS [47] are added for supervising the matting head. The loss function applied are specified in Section 3.2. *Stage 3.* Finally, we fine-tune the model with image matting data D646 [35] and AIM [23] for finer matting details.

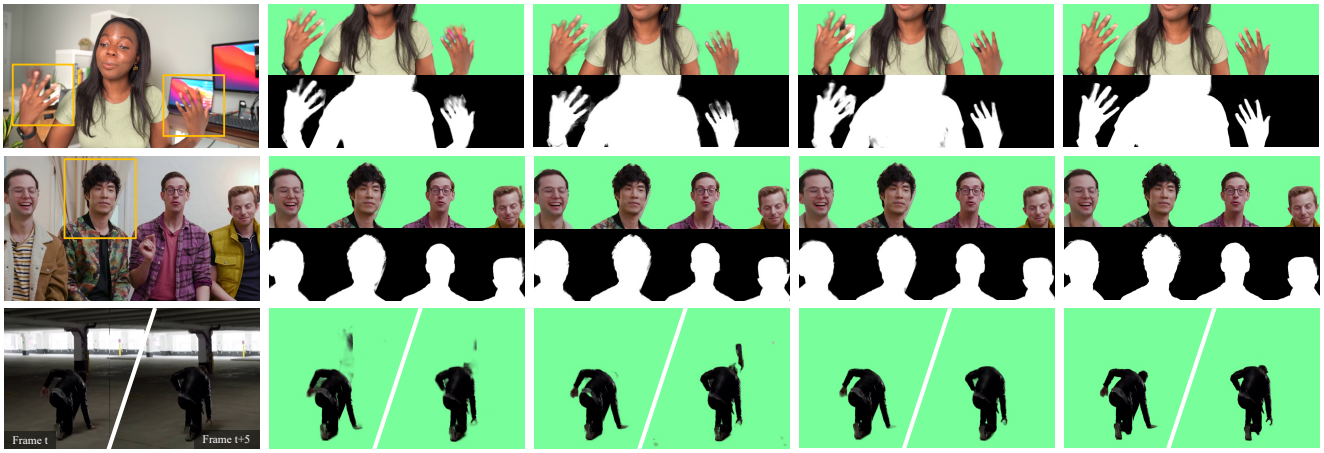


Figure 4. Qualitative comparisons on real-world videos. Our MatAnyone significantly outperforms existing auxiliary-free (RVM [30]) and mask-guided (FTP-VM [19] and MaGGIe [20]) approaches in both detail extraction and semantic accuracy. For the lowest row, while other methods all miss out on important body parts (*i.e.*, head) and mistakenly take background pixels as foreground (due to similar colors), thus generating messy outputs, our method presents an accurate and visually clean output by even identifying the shadow near the boundary.

5.1. Comparisons

We compare MatAnyone with several state-of-the-art methods, including auxiliary-free (AF) methods: MODNet [21], RVM [30], and RVM-Large [30], and mask-guided methods: AdaM [28], FTP-VM [19], and MaGGIe [20].

Table 2. Quantitative comparisons on real-world benchmark [30]. The best and second performances are marked in red and orange, respectively.

Methods	MAD↓	MSE↓	dtSSD↓
Auxiliary-free			
MODNet [21]	11.67	10.12	3.37
RVM [30]	1.21	0.77	1.43
RVM-Large [30]	0.95	0.50	1.30
Mask-guided			
FTP-VM [19]	4.77	4.11	1.68
MaGGIe [20]	1.94	1.53	1.63
MatAnyone (Ours)	0.14	0.10	0.89

5.1.1 Quantitative Evaluations

Synthetic Benchmarks. For a comprehensive evaluation on synthetic benchmarks, we employ MAD (mean absolute difference) and MSE (mean squared error) for semantic accuracy, Grad (spatial gradient) [37] for detail extraction, Conn (connectivity) [37] for perceptual quality, and dtSSD [14] for temporal coherence. In Table 1, our method achieves the best MAD and dtSSD across all datasets at both high and low resolutions, demonstrating exceptional spatial accuracy for alpha mattes and remarkable temporal stability. Apart from accuracy and stability, our method achieves the best Conn on both benchmarks, indicating its superior visual quality (Fig. 4 and Sec. D.5 in the appendix).

Real Benchmark. For evaluation on real benchmarks, we use the core region metrics in Section 4.3. In Table 2, our method demonstrates superior generalizability on real cases, achieving the best metric values with a substantial margin over both auxiliary-free and mask-guided methods.

Table 3. Ablation study of the new training dataset (New Data), consistent memory propagation module (CMP), and new training scheme (New Training) on real benchmark (about 1080p).

Exp.	New Data	CMP	New Training	MAD↓	MSE↓	dtSSD↓
(a)				3.16	2.65	1.37
(b)	✓			2.55	2.25	1.36
(c)	✓	✓		1.85	1.67	1.25
(d)	✓	✓	✓	**0.42**	**0.34**	**0.94**

5.1.2 Qualitative Evaluations

Visual results on real-world videos are in Fig. 4 and Fig. 5.

General Video Matting. MatAnyone outperforms existing auxiliary-free and mask-guided approaches in both detail extraction (*boundary*) and semantic accuracy (*core*). Fig. 4 shows that MatAnyone excels at fine-grained details (*e.g.*, hair in the middle row) and differentiates full human body against complicated or ambiguous backgrounds when foreground and background colors are similar (*e.g.*, last row).

Instance Video Matting. The assignment of target object at the first frame gives us flexibility for instance video matting. In Fig. 5, although MaGGIe [20] benefits from using instance masks as guidance for *each* frame, our method demonstrates superior performance in instance video matting, particularly in maintaining object tracking stability and preserving fine-grained details of alpha mattes.

5.2. Ablation Study

Enhancement from New Training Data. In Table 3, by comparing (a) and (b), it is observed that training with new data noticeably improves the semantic performance

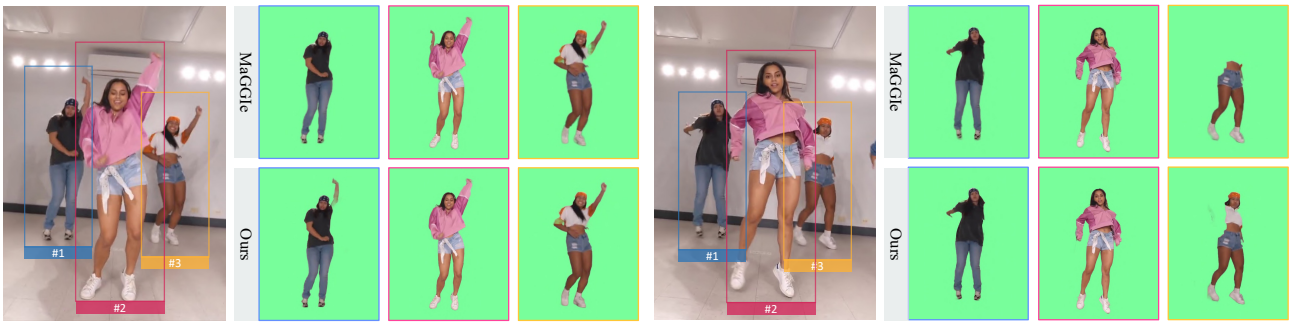


Figure 5. Quantitative comparisons with MaGGIe [20] on instance video matting. Despite MaGGIe using instance mask as guidance for each frame, our method shows better performance, achieving better stability in object tracking and finer alpha matte details.

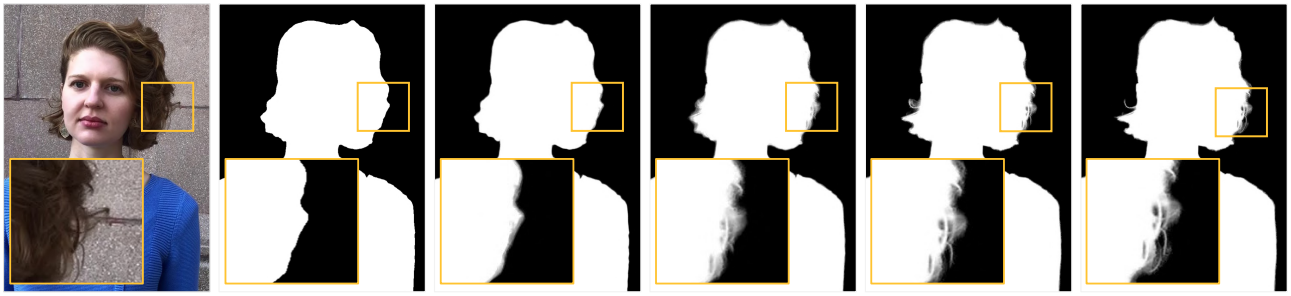


Figure 6. Improvement with Recurrent refinement. **(Zoom-in for best view)**

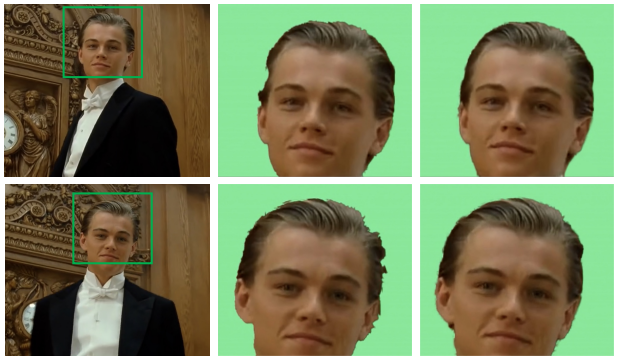


Figure 7. Comparison of matting results training with original DDC loss [32] and with scaled DDC loss, where the latter gives more stable and natural matting results.

with decreased MAD and MSE, showing that our newly-collected VM800 indeed contributes to robust training with its upgraded quantity, quality, and diversity.

Effectiveness of Consistent Memory Propagation. We further investigate the effectiveness of the consistent memory propagation (CMP) module. From Table 3 (b) to (c), improvement can be seen across all metrics with CMP added, indicating its effectiveness in improving semantic stability and temporal coherency. In particular, dtSSD in (c) is already lower than all the other methods in Table 2, showing the superiority of CMP in terms of temporal consistency.

Effectiveness of New Training Scheme. Our new training scheme brings our model to the next level with a noticeable improvement in all metrics. It already outperforms all the other methods in Table 2 without further fine-tuning.

Scaled DDC Loss. We examine the merit of the scaled version of DDC loss by training with $\mathcal{L}_{core}$ and $\mathcal{L}_{boundary}$ only to maximize its effect. In Fig. 7, training with vanilla DDC loss produces segmentation-like jaggedness, especially among the boundary region. Our scaled DDC loss yields more stable and natural matting results.

Effectiveness of Recurrent Refinement. Fig. 6 shows the effectiveness of recurrent refinement in a progressive manner. Given a rough segmentation mask, our method can produce alpha matte with descent details within 10 iterations.

6. Conclusion

We introduce MatAnyone, a practical framework for target-assigned human video matting that ensures stable and accurate results across diverse real-world scenarios. Our method leverages a region-adaptive memory fusion approach, which combines memory from previous frames to maintain semantic consistency in *core* areas while preserving fine details along object *boundaries*. With a new training dataset that is larger, high-quality, and diverse and a novel training strategy that effectively leverages segmentation data, MatAnyone achieves robust and stable matting performance, even with complex backgrounds. These advancements position MatAnyone a practical solution for real-world video matting, also setting a solid foundation for future research in memory-based video processing.

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